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Bringing Sustainability to Operations by Deploying Coated Continuous Rod to Extend Tubing & Rod Run Life in Highly Corrosive Applications

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Abstract

Historically, corrosion has become a challenging condition for oilfield operations when selecting a robust yet reliable Artificial Lift Systems (ALS). Continuous Rod revolutionized the industry during the 2000's, by mitigating tubing and rod wear effect in deviated wells. However, this technology was not a solution in a highly corrosive environment. This paper shows the results of a sustainable project that saved more than fifteen workover interventions and 504 tons of greenhouse gases (GHG) through a coated continuous rod deployed in highly corrosive heavy oil wells. An innovative solution is presented to decrease interventions in wells with rod-driven ALS (extending run life by reducing failures from rod-tubing wear and friction-corrosion) by implementing coated continuous rod technology, avoiding the capital expenditure (CAPEX) for specialized installation equipment. Additionally, this document presents all the benefits (economical and environmental) obtained because of implementing the technology in three wells in operating fields in Colombia.

One of the challenges in implementing recent technology is the cost of installation. The proposed solution was to deliver the product prefabricated from manufacturing (upper rod string coated with epoxy, with welded connector pins and a gooseneck attached to the string under the same conditions and with predetermined/agreed lengths). This approach minimized implementation costs and, in parallel, reduced the operational time at the well site. It also contributed to enhancing the sustainability of the technology under evaluation, demonstrating the benefits that a double-layer, fusion-bonded coating shield on the continuous rod string could provide to three deviated wells.

This article compiles the positive results obtained after correctly transporting the rod string and gooseneck on the same reel from the manufacturing plant to the well location, adapting the available well service units in the field for the correct installation of the technology, and meeting the requirements of the established new technologies protocol.

The document aims to present the optimization process implemented that allowed for a perfect scenario to develop a technological application, starting with the technical process, economic analysis, and the respective benefits obtained in three pilot wells after implementation. Key performance indicators (KPIs)

such as Average Run Life, Sucker Rod Failure, and Intervention Frequency were established, showing excellent results after 3 years of operation without any failures in the ALS.

Keywords: Carbon Emissions, Sustainability, Coated Continuous Rods, Corrosive Fluid, Deviated Wells - Rod and tubing wear, Artificial Lift Systems - ALS

Introduction

La Cira Infantas field selected for the trial is located onshore in the Middle Magdalena basin, near the city of Barrancabermeja, Santander - Colombia. Recovery in the field is through large-scale water injection. Ecopetrol in partnership with SierraCol Energy operate the field. (Chaparro et al, 2019)

The field selected for this technological trial presents characteristics that pose a challenge for rod-driven ALS, which can be summarized in three points: Corrosive fluids, well geometry, and high-water cuts with solids contribution.

Fluid Description

The produced fluid properties in the field are identified in Table 1.

Table 1—Fluid Properties. (Fluid Analysis. 2023).

Well Parameter	Value	Unit
Gas-Oil Ratio	250	scf/STB
Water Cut (WC)	80.5	%
API Gravity	21	°API
CO2 Content	60	ppm
H2S Content	2	%
Bottom Hole Temperature	120	°F
Static Pressure	1200	PSI
Bubble Point Pressure	253	PSI
Formation Volume Factor	1.0501	bbt/STB

The components of this fluid, under the conditions described in the field's general overview, present a challenge that affects all types of equipment in direct contact with the production fluid: corrosion. This material degradation, caused by chemical reactions within the operating environment, is one of the primary causes of failure in the field. Although the use of corrosion inhibitors has been a key focus, the operational characteristics of Artificial Lift Systems (ALS) limit the effectiveness of such treatments, as Progressing cavity pumps (PCP) are developed for sand production but is a problem for chemical treatment because solids particles remove the chemical layer. As an example, Fig. 2 shows one of the most severe failures observed in the field, with a run life of only 52 days in one of the pilot wells selected for this technology trial.



Figure 1—Continuous Rod with severe corrosion (run life of 52 days). (Weatherford Failure Reports, 2022).

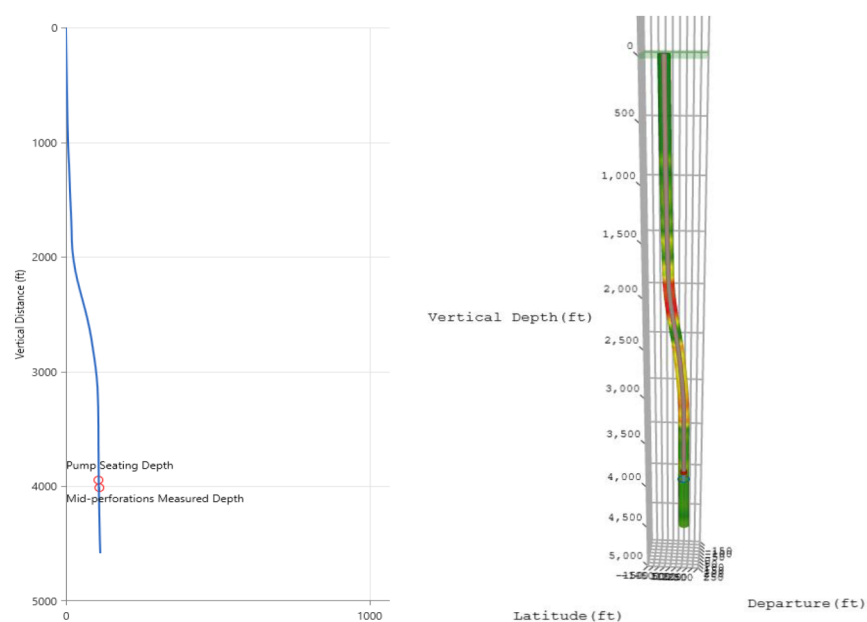


Figure 2—Well geometry vs vertical section [ft] & in three-dimensional view. (C-FER PC-Pump V5.1 2024)

Well Geometry / Construction

Most deviated wells in this field begin with a vertical section, followed by the kickoff point (KOP) at a measured depth (MD) of 500 to 900 ft. In the next angle-building section, the wells reach their maximum angle, which varies from 0° to 10° . For a typical well, Fig. 2 shows the well geometry in terms of a vertical section and a three-dimensional view. Fig. 3 shows the MD vs. DogLeg Severity (DLS), where the typical range in the pay zone is from 2 to $17^{\circ}/100$ ft. The wells are completed with 7 in. production casing from the surface to the bottom or a rat hole length for sand production. The combination of well angle change and aggressive DLS values in an "S"-type well creates a challenging scenario for rod-driven ALS.

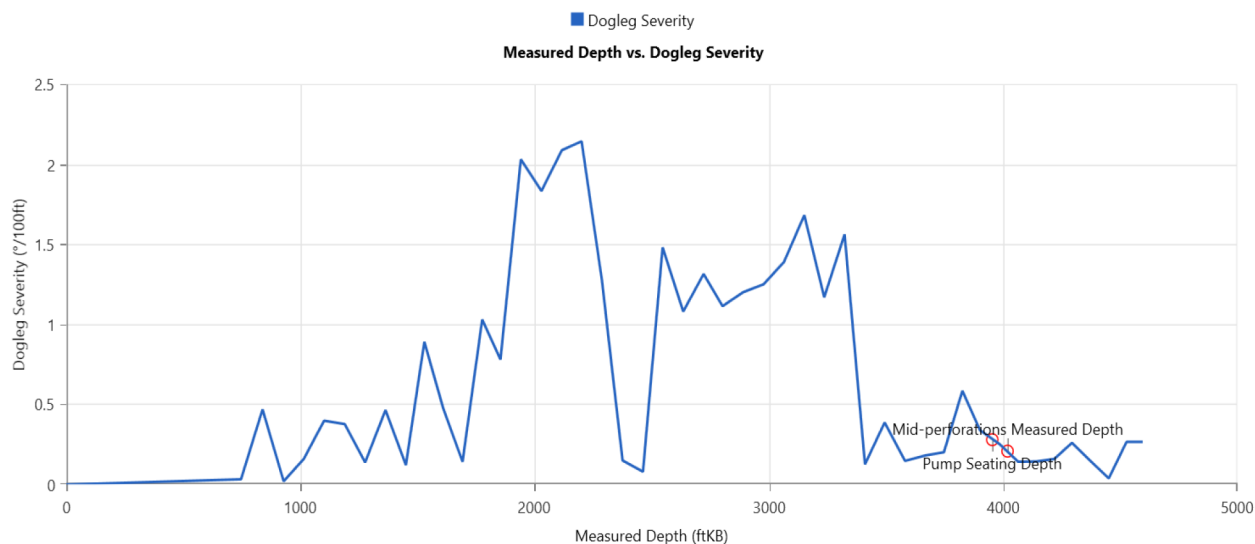


Figure 3—MD vs. DLS. (C-FER PC-Pump. V5.1 2024)

Considering the geometry of the wells in this field, the use of rod-driven ALS becomes a potential for tubing failure due to the wear effect generated by the rod string. This effect can become critical (and in extreme cases, result in tubing broken) depending on several factors, such as the type of fluid produced, the operating regime of the selected ALS, and even the type of sucker rod string used.

The type of sucker rod string can wear out the tubing for several reasons: Due to the continuous movement, whether rotational or reciprocating, of the rod. The second reason is friction with the tubing joints: the contact load between the rod and the tube produces frictional wear on both elements. (Dove et al. 2016)

Contact loads consist of gravity-induced loads and those generated by the curvature of the well (See Fig. 4).

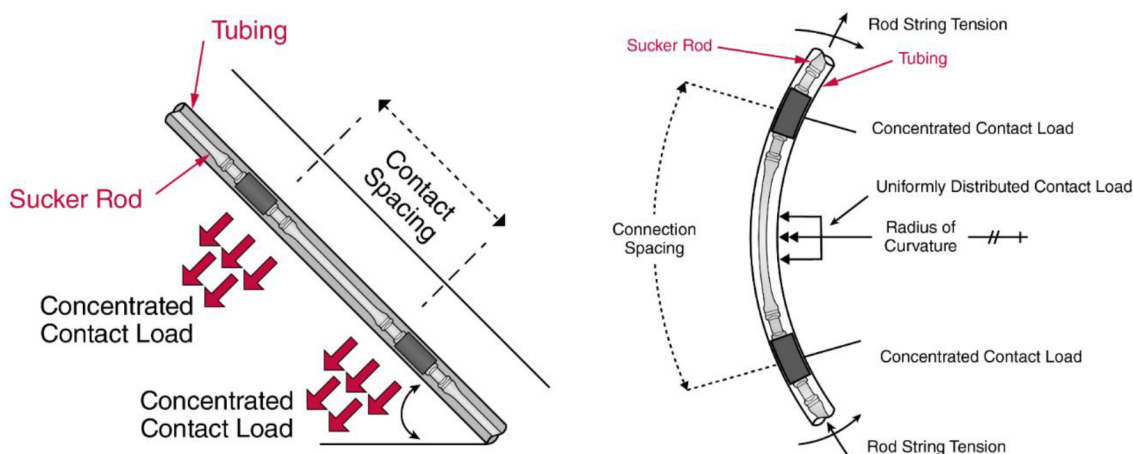


Figure 4—Contact loads induced by gravity and curvature. (Weatherford TSS Corod Data Base. 2014)

The high concentrated contact load specifically at the couplings of conventional rods can be redistributed to uniformly lower values by using a continuous rod. Unlike conventional sucker rods, which are coupled every 25 or 30 ft (7.6 or 9.1 m), the continuous rod has the advantage of operating with only two coupling connections, one at each end of the string. This product with uniformly distributed contact loads makes it ideal for highly deviated or horizontal wells because the rod stresses are distributed along the body. Fig. 6 shows a comparison between the continuous rod and the conventional rod:

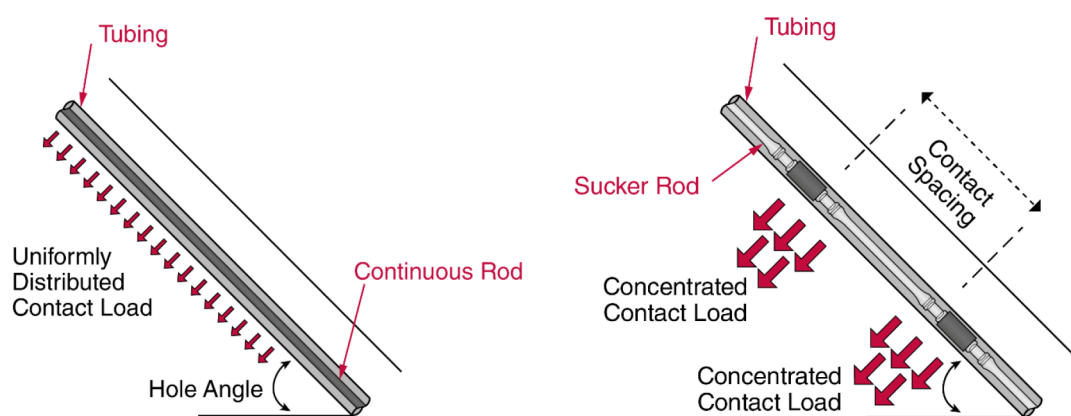


Figure 5—Continuous Rod vs. Conventional Rod. (Tharwat et al. 2023)

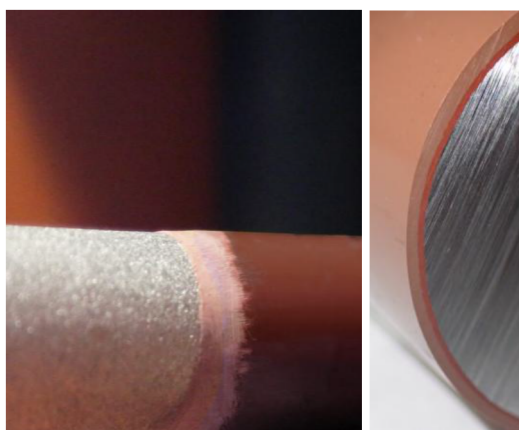


Figure 6—Coated continuous rod with a double-layer coating. (TSS Calabar COROD® Weatherford Data Base. 2018).

The coated continuous rod utilizes an advanced two-stage epoxy coating technology that significantly minimizes the base metal's exposure to corrosive well fluids while reducing wear resulting from contact between the rod and production tubing, due to the coating's low friction coefficient. This solution is particularly suited for rod-driven ALS in wells with a history of corrosion-related failures, as it extends ALS run life by mitigating direct corrosive fluid attack on the rod. Consequently, this reduces operational expenses and production downtime. The coated continuous rod incorporates a fusion-bonded double-layer coating, forming an ultra-thin 0.022-inch protective barrier that prevents direct contact of the base metal with corrosive substances. (Weatherford, 2018)

Water Cut and Solids (Sand) Contribution / Affection

One of the major accelerators of well failures is sand production. Once solids production occurs, preventive measures must be taken, such as reservoir management to reduce the flow rate and thus the solids contribution; or optionally, to produce fluid and sand simultaneously. Usually, the goal is to produce these solids to avoid losing the field's productivity potential, which turns the fluid into an abrasive material. Counterproductively, this directly affects the action of the corrosion inhibitor, as these particles prevent the proper adhesion of the protective film to the materials. When deciding to produce solid particles, the correct choice of ALS is PCP, as it performs better under this condition. However, being a rotary system, its speed and operating cycles accelerate the downhole components failure when combined with an abrasive fluid (transport of suspended solids), which also prevents the proper adhesion of the inhibitor.

Table 2 shows the average solid contribution characteristics of the candidate wells. Fig. 6 shows the coated continuous rod used in this project:

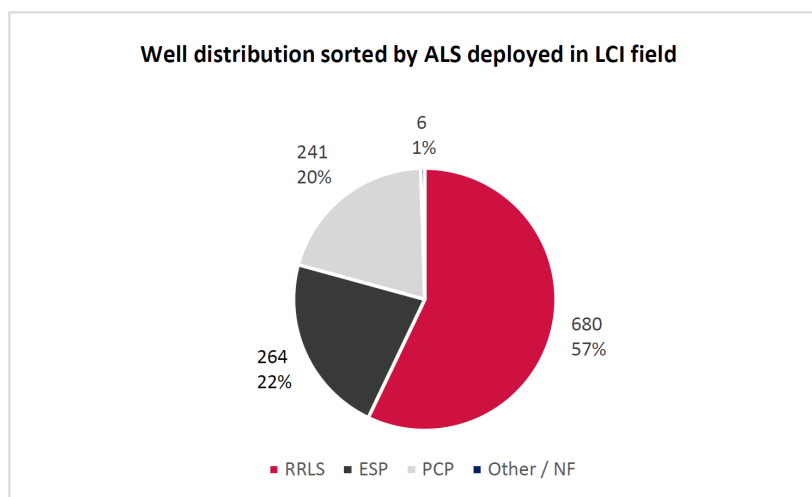
Table 2—Data follow-up to well conditions (Weatherford Failure Reports, 2022).

WBPD	OBPD	FBPD	WC	Average PPM Solid	Max PPM Solid	% Particles between 25–45 microns	% Particles between 45–106 microns	% Particles between 106–250 microns	% Particles between 250–425 microns	% Particles over 425 microns
384.11	79.43	446.92	0.8	59.6276	233.8597	38.2178	32.4938	26.7913	12.9602	8.0133

Methodology

The proposed technology is a ready-to-deploy solution consisting of a fully welded and coated pin-to-pin continuous rod string. Its preassembled delivery significantly reduced implementation costs and eliminated uncertainties associated with field welding of coated continuous rods. Health, Safety, Security, and Environmental (HSSE) considerations were addressed through a formal Management of Change (MOC) process for the new installation, which also emphasized the advantages of the dual-layer, fusion-bonded coating applied to the continuous rod in three critical wells. Well selection was based on failure frequency of rod strings due to wear and corrosion, as well as tubing failures associated with rod/tubing contact and mechanical loading.

The field has about 1191 wells, where more than 99% use ALS to improve fluid production and increase oil recovery rates. Approximately 57% of the wells use reciprocating rod pumping (RRP) systems, 22% are wells with Electrical Submersible Pumping (ESP) systems, 20% use progressing cavity pumping (PCP) systems, while active wells with other ALS and in natural flow represent less than 1% of the total wells. (See details in Fig 8). The coated continuous rod technology is suitable for highly deviated, rod-driven ALS wells (systems with RRP and PCP). The field has a potential of more than nine hundred rod-driven ALS wells that can be reviewed to select candidates for the implementation of coated continuous rod technology, with the aim of reducing the occurrence of failures due to corrosion and friction-corrosion. However, the technology cost and the current failure rate do not imply the need for mass installation in the field. Nevertheless, at least 25% of the wells mentioned have repeated rod failures related to corrosion, friction-corrosion, and/or tubing wear.

Figure 7—ALS systems installed in the field (SierraCol Energy 2023^[11]).

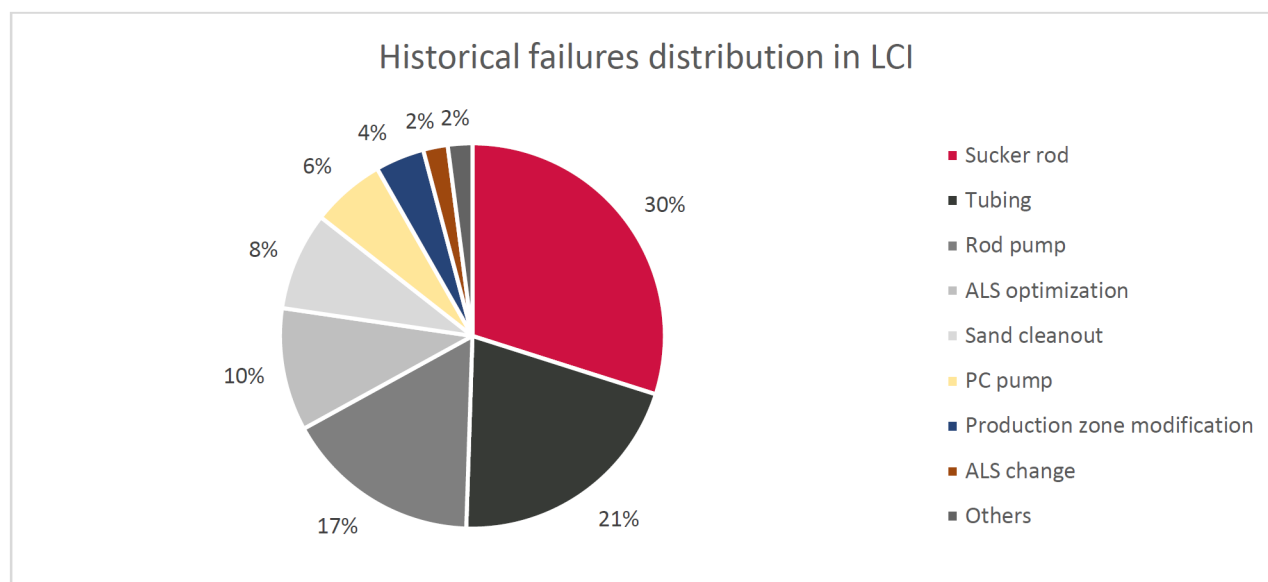


Figure 8—Historical distribution of failures in the field (Weatherford, [SierraCol Failed Wells Database](#). 2022).

The Challenges of Using Rod-Driven ALS in Aggressive Well Geometries

One of the main challenges of using rod-driven ALS (either RRP or PCP) in aggressive well geometries is the frequent failure of rods and tubing due to high well deviation and DLS, which increases the likelihood of rods being affected and affecting the tubing. In addition to the alternating operation of the rod string, these contact points can lead to accelerated or even severe tubing wear. The final wear rate and the run life of the rod depend on factors such as DLS, the final well angle, pumping speed, and corrosion. The greater the wear, in this case, the higher the OPEX (replacement of tubing and rods) and the well's downtime, resulting in a loss of revenue by deferring oil production. Furthermore, frequent rod failures result in complicated fishing jobs and workover operations to restore the well's production.

Causes of Failures

According to field information shown in [Fig. 8](#), about 68% of well failures are related to downhole equipment. Rod string failures are around 29% and tubing failures correspond to 20% of the cause of failure in the wells. Well service interventions due to tubing are considered one of the most expensive problems to address; in most cases, this kind of service intervention will require to pull out of hole (POOH) the well completion (POOH), complete the well service and restore production in the asset. As result, a workover rig will be needed at the well site, but the waiting time for this type of intervention could be days or even weeks due to the lack of availability of this kind of units in the area. In Colombia, the waiting time due to the availability of well workover equipment is longer for low-potential wells, as the field service units are most often assigned to emergencies with high-production rate wells and better Water Cut (WC).

In addition to the failure mechanism due to well geometry, there is the chemical component factor, which is related to the production fluid and its effects on the well completion elements. A catalyst to the condition is the loss of reservoir potential, which requires resorting to enhanced recovery methods to optimize crude oil extraction. The initial Enhance Oil Recovery (EOR) method for this field is water flooding, which aims to improve oil swept within the producing formation. Consequently, these fluids accelerate corrosive effects on the ALS components; corrosive effects are difficult to control in dynamic systems, and even more in conditions of impure fluids with the presence of solids.

Based on the previously mentioned criteria, the well selection process for the technology trial was conducted. Wells with the poorest performance were chosen, based on indicators such as rod and tubing

failures, particularly those primarily affected by corrosion and showing low effectiveness of chemical treatment injections.

Below are the failure statistics basis for the selection of the pilot wells:

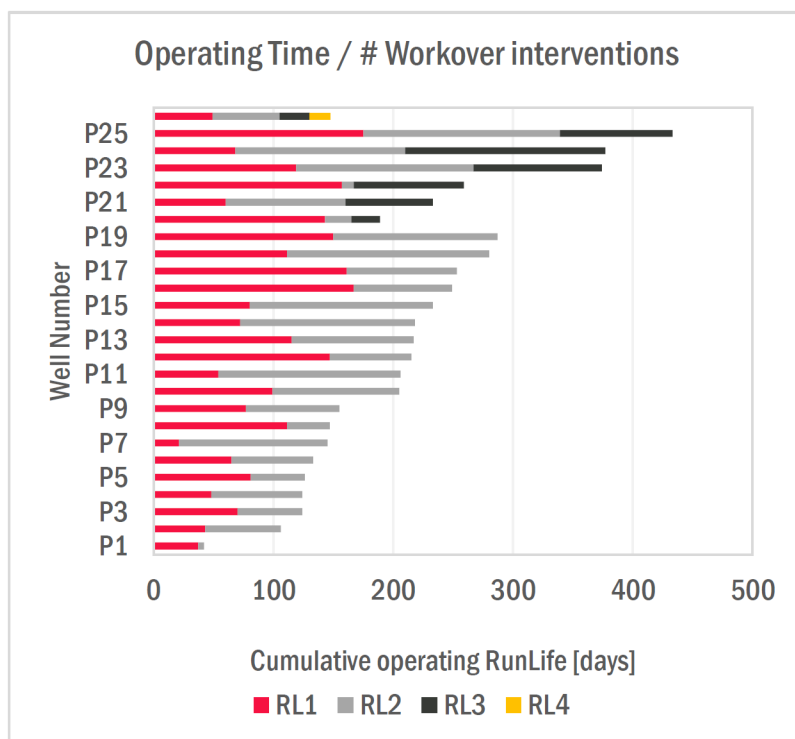


Figure 9—Historical distribution of failures in the field (Weatherford, [SierraCol Failed Wells Database. 2022](#)).

Once the potential candidates were selected and the client completed the internal economic evaluation, the intervention was planned. The conditions for installing the technology were defined, requiring certain adaptations to the Flush By unit that would deploy the coated rod. One of the requirements included placing a trailer at the wellhead to carry out the welding and application of the specialized coating. This process involved importing high-cost equipment, which typically requires a high volume of operations to justify the expense; however, the technology trial was limited to only three wells, in accordance with the client's protocol.

Given the urgency to deploy the technology, the client expanded the scope to include additional wells with similar conditions, increasing the likelihood of installation as soon as a failure occurred. The critical condition for implementation remained unchanged. However, the CAPEX associated with importing specialized welding equipment proved cost-prohibitive, as it had to be absorbed by one of the parties, making the original plan unviable. To address this, the Weatherford ALS team proposed a more efficient alternative: manufacturing a factory-finished, pin-to-pin rod string of a specific length. This solution eliminated the need for on-site welding and significantly reduced import costs, restoring the project's financial feasibility. It's important to note that this approach limited well selection to a predefined depth range.

With the approval of this innovative way of performing the well intervention, the Flush By units were modified and adapted to be able to deploy both types of rods (bare continuous rod and coated continuous rod, respectively), reducing downtime for preparation.

These modifications involved changing the pads of the main chain of the continuous rod injector, as the gripping surface would not be the usual one (Steel). For this technology, a change in the groove pattern of these pads were required, a modification that would increase the string's resistance to slipping and reduce the pressure required by the injector to run the rod inside the tubing. Additionally, centralizers or rollers

were used to keep the product inside the continuous rod guide, without it touching the walls and having the possibility of deterioration before starting its run inside the well. Finally, the safety brake clamp was modified, which also contained the required pattern to not damage the string in case of an emergency.

This logistical alignment between companies required several technical sessions to gain a better understanding of the concept and to make the installation of the technology feasible. Once the modifications were finalized, the first candidate well, which had experienced a premature rod failure due to corrosion, was selected for intervention. The PCP system was chosen for the three pilot wells, as this type of rod-driven ALS typically has a higher failure rate. The chemical treatment for corrosion inhibition tends to be less effective in these systems. Additionally, this type of ALS handles larger volumes of solids and experiences greater wear due to rod/tubing contact.

Greenhouse Gas Emissions

The globally recognized methodology for calculating GHG emissions is the GHG Protocol. This protocol provides a framework for organizations to consistently and transparently account for and manage their GHG emissions. The main key steps for calculating emissions according to the GHG protocol:

- A. Define accounting boundaries, determining the organizational and operational boundaries to establish the scope of the emissions measurement.
- B. Identify emission sources: Pinpoint all relevant GHG sources within the defined boundaries, including direct emissions related to the project (Scope 1), indirect energy emissions (Scope 2), and other indirect emissions (Scope 3).
- C. Collect activity data: Gather data on activities that generate emissions, such as fuel consumption, electricity use, transportation, and more.
- D. Apply emission factors: use emission factors to convert the activity data into GHG emissions. Emission factors are values that link an activity to the quantity of GHG emitted (e.g., kg of CO₂ per kWh of electricity).
- E. Calculate emissions: Calculate emissions for each source and then sum them up to get the organization's total GHG emissions.
- F. Develop an emissions report: Create a detailed report that includes the accounting boundaries, emission sources, collected data, emission factors, and calculated emissions.

Other considerations:

- Scope 1: This includes direct emissions from sources controlled by the organization, such as fuels burned in owned facilities, company vehicles, etc.
- Scope 2: This includes indirect emissions associated with the electricity, heat, or steam consumed by the organization.
- Emission factors: These are values applied to activities to calculate GHG emissions and are available in various databases and standards.
- Units of measurement: Emissions are typically expressed in metric tons of CO₂ equivalent (tCO₂e).

GHG Protocol is the most widely used methodology for calculating GHG emissions. As described by [Eurofins Environment Testing Spain](https://onepetro.org/SPEADIP/25ADIP/conference/25ADIP), its application involves defining boundaries, identifying sources, collecting data, using emission factors, and calculating total emissions, with the goal of producing a transparent and accurate report.

For the oil industry, the emissions calculations results obtained through this calculator are estimates based on emissions factors published by the American Petroleum Institute (API) and rig engines specifications (power rating, number of engines, etc.,) are typical and align with a drillship.

Emission Factors obtained from the [API 2023](#) Compendium of Greenhouse Gas Emissions Methodologies for the Natural Gas and Oil Industry (page 707). The emission factor assumes a load factor of 75% in accordance with the API Compendium.

Results

A total of three fit-for-purpose, pin-to-pin coated continuous rod strings were transported from the manufacturing facility to the well sites of the selected candidate wells. The implementation of the coated continuous rod extended the run life of these wells by more than 4.9 times compared to previous sucker rod systems. This performance exceeded the operator's expectations, achieving over three years of operation without failures or workover interventions related to broken sucker rods or tubing wear.

A reduction of 81% in the total expenses was achieved with the implementation of this custom-made deployment, improving rod string run life up to 741%. Number of workover interventions per year, and annual number of sucker rod failures were the KPIs set for this field trial. This technology is by far a feasible, highly sustainable, and proven solution for wells historically affected by corrosive fluids originating multiple sucker-rod-broken interventions; the tubing-rod contact was also drastically reduced in these wells. The project brought 750K USD savings per well. This is equivalent to a reduction of over 1.3 MMUSD in workover interventions and over five hundred tons of GHG for this project.

The market for this technology is high failure frequency wells, where technology implementation can reduce the carbon footprint generated by frequent workover equipment and rigs, specialized flush by units, transportation, lift equipment, pickup trucks, among others. Once the expectation of average failure time changes from 200 days up to 1000 days or more (today's run life), GHG savings up to 162 Tons (per well): 3 years with no rig interventions.

When comparing the pilot wells executed with this project and their performance results, an analysis of the intermediate values of the three wells was conducted to evaluate their behavior compared to other rods.

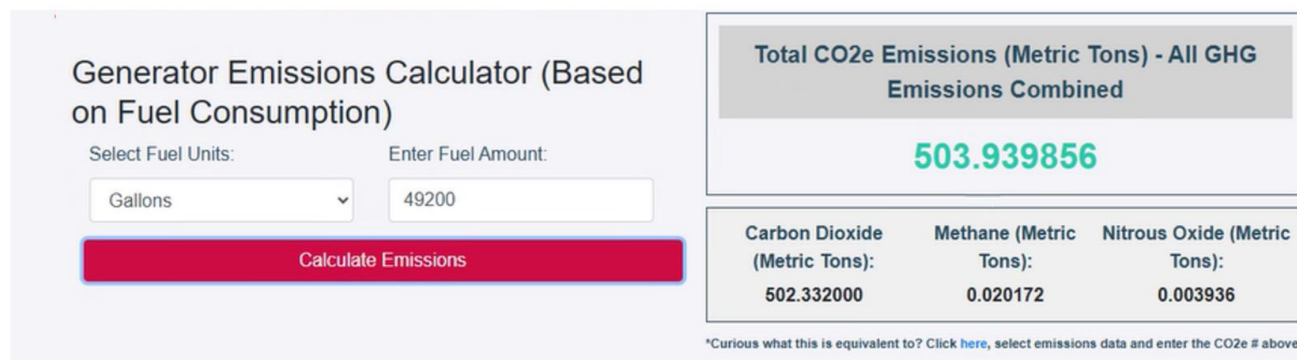


Figure 10—GHG Weatherford calculator for emissions generated by diesel consumption in different equipment. (Generator Emissions calculator, Weatherford)

This is equivalent to greenhouse gas emissions from:



This is equivalent to CO₂ emissions from:



Figure 11—Equivalent comparison of emission of GHG saved with coated continuous rod technology implementation. (Environment protection agency Greenhouse Gas Equivalencies Calculator (EPA))

To visualize the product's performance, a comparison was made only with the components replaced when deploying this technology, which is the rod. Although technology also improves the friction conditions and the performance of the tubing or a diagnosis for a broken tubing, the comparison is limited to the replacement time of the rod. The following graph shows the changes or replacements of the rod in the well.

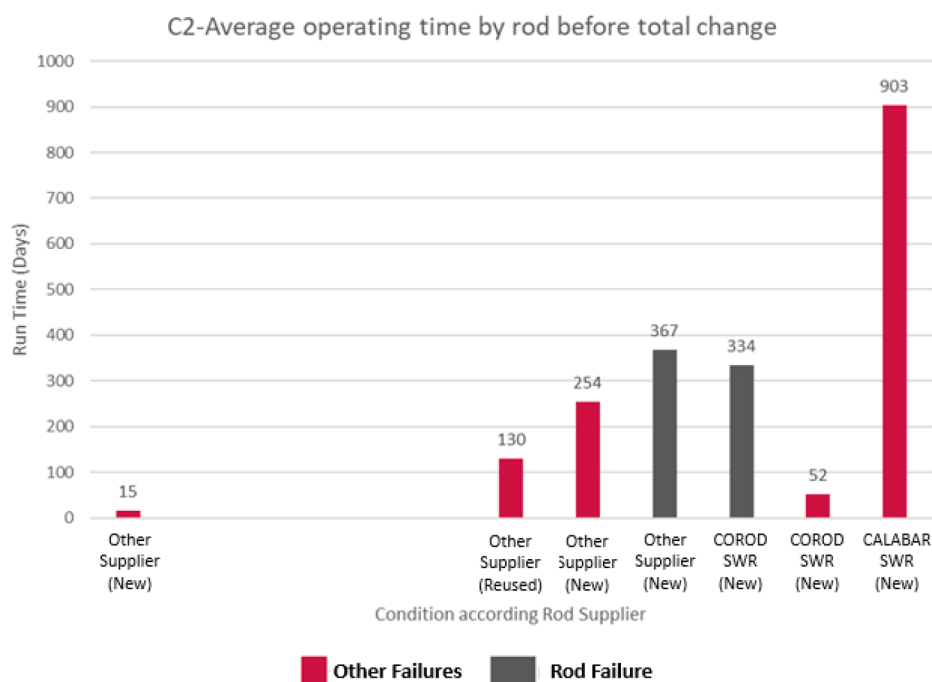


Figure 12—Statistics of average run life in Calabar pilot wells. (Weatherford, SierraCol Failed Wells Database. 2022).

In the well's operational history, the maximum run time of the rod did not exceed 367 days. With the implementation of the coated continuous rod, performance reached 903 days, surpassing the previous best run life by 2.46 times. From an economic standpoint, at least two well interventions were avoided, along with

associated costs such as non-productive time during interventions (including equipment standby), material consumption (e.g., new rod strings), and labor hours required to execute these operations.

This successful implementation highlights the strong commitment of all stakeholders to continuous process improvement, as well as the dedication of the technology proponents to research, development, and field validation. It also underscores the willingness to assume operational risk in pursuit of meaningful performance gains at the asset level.

Discussion

The performance of the coated continuous rod was better than expected, as the protocol established to declare successful the field trial was set to 180 days of operation without rod failures. At the time this article was written, the wells have been operating for more than 2 years without reporting any failure or discontinuity in their operation. Based on this, the continuous operation time of each of the wells with this technology was multiplied by an average of 3.47X.

The historical average of failures associated with the ALS and classified as rod failure for these wells, prior to the installation of the technology, was less than 200 days in the most critical wells C1 and C2, and 373 days for well C3, the most challenging one having better statistical numbers. We are less than seventh months away from tripling the run life in the best performing well of the trial.

Notably, the best run life previously recorded in these well-regardless of failure type, was exceeded by the performance achieved with the technology in all three instances. This successful case provides robust data from a Latin American application, supporting its potential for deployment in other fields with similar characteristics. Both technically and financially, the coated continuous rod demonstrated clear benefits throughout the trial, resulting in a 100% success rate.

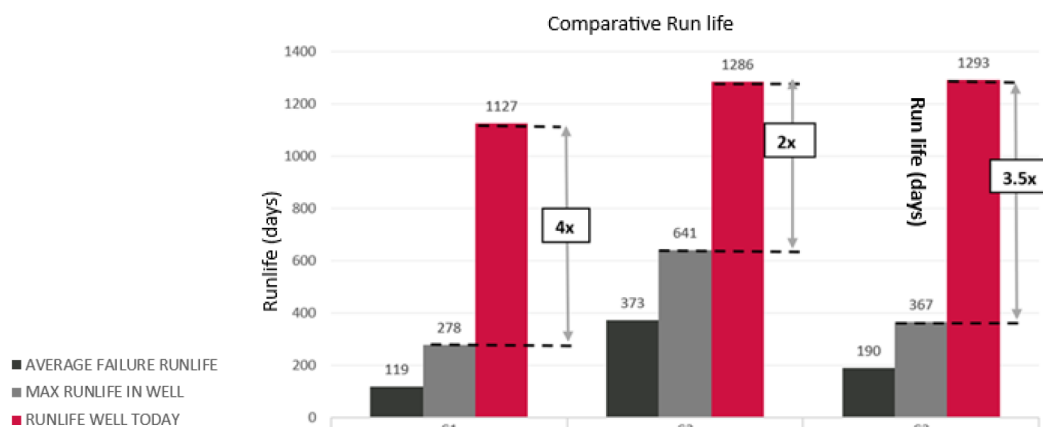


Figure 13—Statistics of average run life in Calabar pilot wells.

One of the project's initial objectives was to analyze the condition of the installed product after a defined run life, specifically to assess wear from rod/tubing contact, the impact of well fluids on the coating, and overall mechanical integrity. However, and fortunately, none of the pilot wells have required a (POOH) operation to date. As a result, the evaluation of the rod condition will take place following the publication of this article.

In the meantime, the failure rate is estimated to have decreased, and remains below 0.33, with all wells operating for over three years without failure.

Conclusions

- The solution offered to install the technology without the need for CAPEX and the import of welding equipment was crucial to proceed with the implementation of a new development feasible in such a way that it did not have a severe economic impact that would compromise its installation. An 81% reduction in total expenses was achieved between the implementation of the welding equipment method versus the methodology used for the execution of this project.
- The evaluation of the performance of the three pilot wells before and after the deployment shows a 348% improvement over the expected run life in case of rod string failure, giving an advantage to this technology as an energy saving solution.
- The rod's performance has demonstrated a correct solution to the usual problems of the field's wells, mitigating production downtime, loss of time due to intervention, reduction of the failure rate, and reduction of investment and costs in well service jobs. The operational continuity of the system with this technology has set a new challenge in the operational performance of the field.
- The simple fact to reduce the number of interventions per year generates a massive impact on GHG, expanding the expectations of emission every 200 days to the same amount through more than 1000 days, saving more than 500 tons of emission in the time that this project has been running. This gives precise value to technology, making it a sustainable solution.
- The number of well service interventions per year and the annual number of sucker rod failures were the KPIs that showed excellent results, with an operational run life 2 times longer than ever recorded in these wells, making it an ideal and proven technology for wells historically affected by interventions for sucker rod failures due to corrosive fluids.
- Coated Continuous Rod technology also drastically reduces the contact between tubing and rod, which is evidenced by the non-occurrence of tubing failures.
- The installation of coated continuous rod strings extended the average run life in these wells by more than 3.47 times compared to other sucker rod products previously implemented, exceeding the client's expectations (more than 2 years without failures).

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Nomenclature

Abbreviation	Full Term
ALS	Artificial Lift Systems
CAPEX	Capital Expenses
COROD	Continuous Rod
DLS	Dogleg Severity
EOR	Enhanced Oil Recovery
ESP	Electrical Submersible Pumping
GHG	Greenhouse Gases
HSSE	Health, Safety, Security and Environmental
KPI	Key Performance Indicator
KOP	Kickoff Point

LCI	La Cira Infantas
MD	Measured Depth
OPEX	Operating Expenditure
PCP	Progressing Cavity Pumping
POOH	Pull Out of Hole
RRP	Reciprocating Rod Pumping
USD	US Dollars
Vs	Versus
WC	Water Cut
bbl/STB	Barrel per stock tank barrel
Psi	Pound per square inch
m	Meters
scf/STB	Standard cubic feet per stock tank barrel
°	Degree
%	Percentage
°F	Degree Fahrenheit
CO ₂	Carbon Dioxide

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